
Targeting Human SAH Hydrolase (AHCY) to Suppress MERS-CoV Replication: A Structure-Based Docking Study of 6'-Fluorinated Aristeromycin Analogues

BMEG 250 Term Project

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ABSTRACT

Middle East respiratory syndrome coronavirus (MERS-CoV) caps its genomic RNA using S-adenosylmethionine (SAM)-dependent methyltransferases and encodes no S-adenosylhomocysteine (SAH) hydrolase of its own, so it depends on the host methylation cycle. Inhibiting human SAH hydrolase (AHCY) raises SAH levels, which is expected to product-inhibit methyltransferases and impair viral RNA capping. This study evaluated human AHCY as a target for six aristeromycin-derived compounds reported by Yoon et al. (a parent compound, four 6'-fluorinated analogues and a phosphoramidate prodrug). The 2.01 Å crystal structure 1LI4 was selected as the receptor after a comparison with an AlphaFold model, whose confidence was too low to support docking. SwissDock placed the five active compounds in the substrate pocket adjacent to the NADH cofactor, with best scores of -5.8 to -7.2 kcal/mol (estimated $K_D \approx 5\text{--}60\text{ }\mu\text{M}$), and gave strongly positive scores (+22 to +25 kcal/mol) to the bulky prodrug, consistent with its inactivity against the isolated enzyme. Docking ranked 6',6'-difluoroaristeromycin (2c) highest, but it did not reproduce the experimental ordering of enzyme inhibition (Spearman $\rho = -0.10$, $n = 5$). The results illustrate both the usefulness and the limits of rigid-receptor docking for closely related, mechanism-based inhibitors.

Keywords: SAH hydrolase; AHCY; MERS-CoV; aristeromycin; molecular docking; SwissDock; host-directed antivirals

1. Background and Objective

1.1 SAH hydrolase and the methionine cycle

S-adenosylhomocysteine hydrolase (SAH hydrolase; gene *AHCY*) is a highly conserved enzyme of the methionine cycle. It catalyzes the reversible hydrolysis of SAH to adenosine and homocysteine. Homocysteine can enter the transsulfuration pathway or be remethylated to methionine, which is used to regenerate SAM, the universal methyl donor for cellular methylation reactions. Every SAM-dependent methylation releases SAH, a potent product inhibitor of methyltransferases, so continuous removal of SAH by AHCY is required to keep the SAM:SAH ratio, and therefore cellular methylation capacity, high.

1.2 Rationale for host-directed inhibition

Coronaviruses cap the 5' end of their genomic RNA using SAM-dependent methyltransferases (nsp14 N7-methyltransferase and nsp16 2'-O-methyltransferase). The cap stabilizes viral RNA and helps it avoid innate immune sensing. Inhibiting host AHCY causes SAH to accumulate, which is expected to product-inhibit these methyltransferases and impair capping and replication. Because MERS-CoV does not encode an AHCY homologue, the host enzyme is the relevant target. The 6'-fluorinated aristeromycin analogues studied here were designed as dual-target antivirals directed at both the host SAH hydrolase and the viral RNA-dependent RNA polymerase (RdRp) [1].

1.3 Objective

This project uses molecular docking to (i) select and prepare a suitable structural model of human AH CY, (ii) dock six aristeromycin-derived compounds from Yoon et al. [1] into the substrate-binding pocket, and (iii) compare the predicted binding with the experimentally measured enzyme inhibition (IC_{50}) and antiviral activity (EC_{50}).

2. Methods

2.1 Target selection: viral or human AHCY?

MERS-CoV does not encode its own AHCY, so the virus relies on the host enzyme to recycle SAH and sustain the SAM supply needed for RNA capping. To check whether a viral AHCY-like protein could be a competing target, human AHCY (UniProt P23526) was aligned with an AHCY-like viral sequence from UniProt (A0A502W8X0) using T-Coffee [4]. The alignment showed low consistency across most of the sequence (Fig. 1). AHCY-like sequences annotated in viral records tend to belong to the formate/glycerate dehydrogenase family or to originate from large DNA viruses; they show little structural or functional similarity to the human enzyme and have no known role in MERS-CoV infection. The human enzyme was therefore chosen as the target for all structural and docking work.



Figure 1. T-Coffee multiple sequence alignment of human AH CY (sp|P23526) and a viral AH CY-like sequence (tr|A0A502W8X0). Colors indicate alignment consistency (red = poor, green = good). Poorly conserved regions predominate, supporting the choice of the human enzyme as the docking target.

2.2 Identification of experimental structures

A BLASTp search [3] against the Protein Data Bank was used to retrieve experimentally determined AH CY structures. Human structures (including 1LI4, 4PGF, 4PFJ and 3NJ4) ranked at the top with about 95.7% identity over the aligned region, followed by rat, mouse and more distantly related bacterial and protozoan enzymes (Fig. 2).

Sequences producing significant alignments			Download	Select columns	Show	100			
☒ select all 41 sequences selected			GenPept	Graphics	Distance tree of results	Multiple alignment	MSA Viewer		
	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Chain A, S-adenosylhomocysteine hydrolase [Homo sapiens]	Homo sapiens	140	140	16%	8e-37	95.71%	432	1LI4_A
☒	Chain A, Adenosylhomocysteinease [Homo sapiens]	Homo sapiens	140	140	16%	8e-37	95.71%	432	4PGF_A
☒	Chain A, Adenosylhomocysteinease [Homo sapiens]	Homo sapiens	140	140	16%	9e-37	95.71%	432	4PFJ_A
☒	Chain A, Adenosylhomocysteinease [Homo sapiens]	Homo sapiens	140	140	16%	9e-37	95.71%	435	3NJ4_A
☒	Chain A, Adenosylhomocysteinease [Mus musculus]	Mus musculus	140	140	18%	1e-36	88.31%	432	5AXA_A
☒	Chain A, Adenosylhomocysteinease [Mus musculus]	Mus musculus	140	140	18%	2e-36	88.31%	452	8CQD_A
☒	Chain A, Adenosylhomocysteinease [Rattus norvegicus]	Rattus norvegicus	137	137	18%	2e-35	86.84%	431	1XWF_A
☒	Chain A, S-ADENOSYLHOMOCYSTEINE HYDROLASE [Rattus norvegicus]	Rattus norvegicus	137	137	18%	2e-35	86.84%	431	1D4F_A
☒	Chain A, PROTEIN (S-ADENOSYLHOMOCYSTEINE HYDROLASE) [Rattus norvegicus]	Rattus norvegicus	137	137	18%	2e-35	86.84%	431	1B3R_A
☒	Chain A, Adenosylhomocysteinease [Homo sapiens]	Homo sapiens	134	134	16%	1e-34	95.52%	429	5W49_A
☒	Chain A, S-ADENOSYLHOMOCYSTEINE HYDROLASE [Homo sapiens]	Homo sapiens	128	128	16%	3e-32	89.86%	432	1A7A_A
☒	Chain A, Adenosylhomocysteinease [Leishmania major]	Leishmania major	105	105	16%	6e-24	67.61%	437	3G1U_A
☒	Chain A, Adenosylhomocysteinease [Drosophila melanogaster]	Drosophila melanogaster	105	105	20%	6e-24	62.07%	434	9PDH_A
☒	Chain A, Adenosylhomocysteinease [Elizabethkingia anophelis NUHP1]	Elizabethkingia anophelis NUHP1	102	102	15%	3e-23	69.70%	445	6APH_A
☒	Chain A, Adenosylhomocysteinease [Legionella pneumophila]	Legionella pneumophila	102	102	16%	4e-23	67.61%	437	8WWG_A
☒	Chain A, Adenosylhomocysteinease [Legionella pneumophila]	Legionella pneumophila	102	102	16%	4e-23	69.57%	437	8WZ7_A
☒	Chain A, Adenosylhomocysteinease [Trypanosoma brucei]	Trypanosoma brucei	100	100	16%	1e-22	66.20%	436	3H9U_A
☒	Chain A, Adenosylhomocysteinease [Legionella pneumophila]	Legionella pneumophila	99.4	99.4	16%	5e-22	66.20%	437	8WZ8_A
☒	Chain A, Adenosylhomocysteinease [Burkholderia pseudomallei 1710b]	Burkholderia pseudomallei 1710b	98.6	98.6	16%	1e-21	64.79%	494	3D64_A
☒	Chain A, Adenosylhomocysteinease [Cytophaga hutchinsonii ATCC 33406]	Cytophaga hutchinsonii ATCC 33406	97.1	97.1	14%	3e-21	68.85%	438	6GBN_A

Figure 2. Top BLASTp hits against experimentally determined structures in the PDB.

2.3 Receptor choice: PDB 1LI4, chain A

Chain A of PDB entry 1LI4 [2], an X-ray structure of human AH CY in complex with neplanocin, was selected as the primary receptor for the following reasons:

- **Resolution and quality.** The structure was solved at 2.01 Å with $R_{\text{free}} = 0.236$ and $R_{\text{work}} = 0.197$, giving reliable side-chain coordinates. This matters for docking, where small errors in pocket geometry can produce false positives.
- **Cofactor.** The tightly bound NAD(H) cofactor is present in the cofactor-binding domain. Because AH CY uses this cofactor in its catalytic cycle, retaining it keeps the active-site geometry biologically realistic.
- **Catalytically relevant conformation.** Neplanocin is bound in its 3'-keto form, a substrate analogue and inhibitor, and holds the enzyme in the closed conformation in which catalysis occurs [2]. Removing the ligand leaves a well-defined, closed pocket suited to evaluating new inhibitors.
- **Human origin.** Docking against the human enzyme keeps the analysis tied to the host target that MERS-CoV depends on.

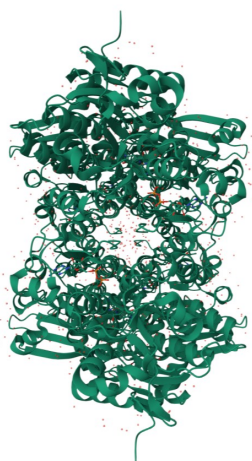


Figure 3. Overview of the experimental crystal structure of human AHCY (PDB 1LI4), shown with bound ligands and crystallographic waters. The deposited entry contains four subunits; chain A was used for docking.

2.4 Comparison with an AlphaFold model

For comparison, a structure prediction was generated with AlphaFold2 [5] through ColabFold [6]. Predicted models can supply full-length coordinates with per-residue confidence (pLDDT), which is useful for regions that are often missing from crystal structures, such as flexible loops and termini. In this run, however, the multiple sequence alignment was deep only over roughly the first 80 residues and sparse beyond about residue 200 (Fig. 4). The resulting models showed low confidence (pLDDT < 50) over most of the structure (Fig. 5). Docking into a low-confidence pocket would yield unreliable affinity estimates, so the experimental structure was used as the docking receptor.

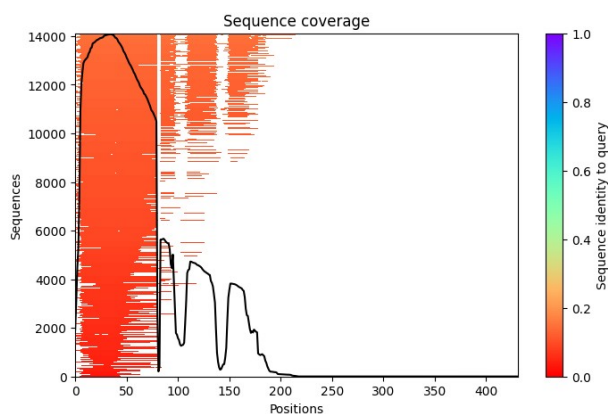


Figure 4. Sequence coverage of the ColabFold multiple sequence alignment across the query (≈430 residues). Depth is high only for the N-terminal region.

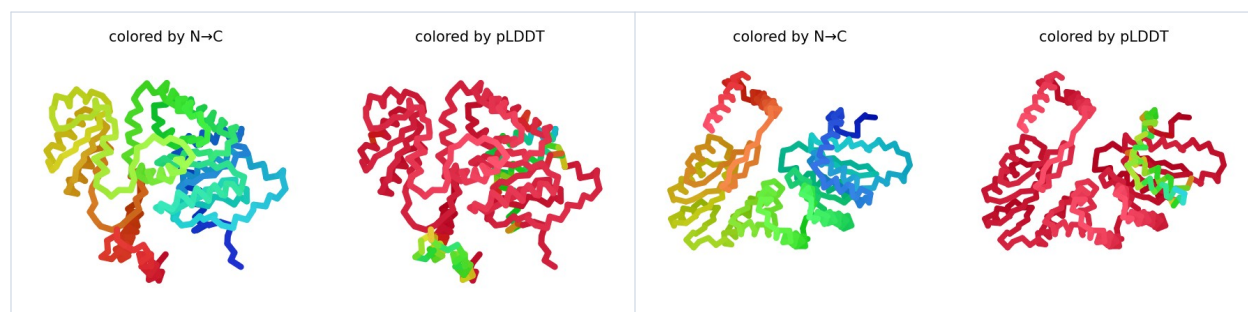


Figure 5. Two ColabFold models colored by chain position (N → C, left panels) and by pLDDT (right panels). The predominantly low-confidence coloring on the right is the reason these models were not used for docking.

2.5 Receptor preparation

Chain A of 1LI4 was isolated and the bound neplanocin was removed in PyMOL [8] to expose the substrate-binding pocket. The NADH cofactor was retained to preserve the native catalytic geometry.

2.6 Active site and docking box

The substrate-binding pocket lies in the deep cleft between the substrate- and cofactor-binding domains. Residues that contact the substrate were compiled from the structural literature on SAH hydrolase (Table 1, Fig. 6). The docking box was centered on the empty substrate-binding cavity next to the retained NADH, at the geometric center of these residues: $x = 45.35 \text{ \AA}$, $y = -18.36 \text{ \AA}$, $z = 106.33 \text{ \AA}$.

Table 1. Active-site residues used to define the docking region.

Interaction	Residues
Ribose binding	His55, Asp131, Glu156, Thr157, Lys186, Asp190, His301
Adenine binding (hydrogen bonds)	Thr57, Glu59, main-chain atoms of His353
Adenine binding (hydrophobic pocket)	Leu54, Leu347, Met358, Phe362

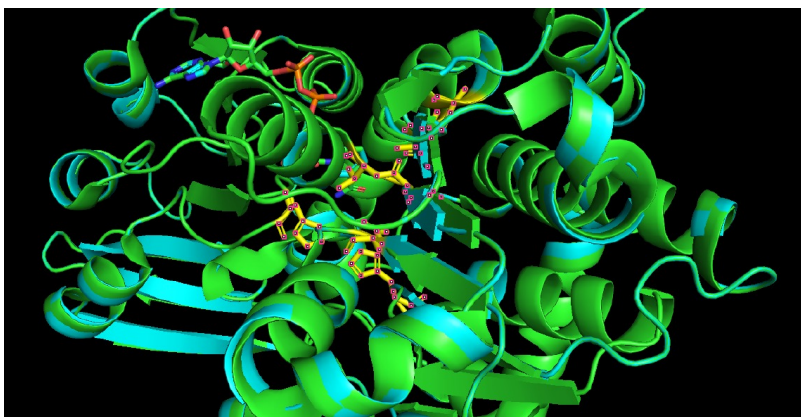


Figure 6. Active-site residues (yellow sticks) mapped onto the AHCY structure in PyMOL.

2.7 Ligand set

Six compounds were taken from the medicinal chemistry study of Yoon et al. [1] (Table 2). They span a range of enzyme-inhibition and antiviral profiles, which makes them a useful test of whether docking scores track biological activity. The parent compound provides the baseline, 2d serves as an antivirally inactive comparator, and prodrug 3a is expected to be excluded from the pocket.

Table 2. Ligands selected for docking.

ID	Compound	Role in this study
1	(-)-Aristeromycin	Parent compound; baseline for all fluorinated analogues
2a	6'-β-Fluoroaristeromycin	Most potent SAH hydrolase inhibitor in the series
2c	6',6'-Difluoroaristeromycin	Most promising broad-spectrum antiviral candidate; strong MERS-CoV inhibition with lower cytotoxicity
2d	N ⁶ -Methyl-6'-β-fluoroaristeromycin	Antivirally inactive comparator (MERS-CoV EC ₅₀ > 50 μM); expected to dock poorly
2e	N ⁶ -Methyl-6',6'-difluoroaristeromycin	"Middle-ground" compound: sub-micromolar enzyme inhibition but only moderate antiviral activity
3a	Phosphoramidate prodrug of 2c	Bulky prodrug, inactive against the isolated enzyme; expected to clash with the pocket

Experimental values for all compounds (Table 3) are taken from Yoon et al. [1].

2.8 Docking and conversion of scores to K_D

Each ligand was docked into the prepared receptor with SwissDock [7]. SwissDock returns a ranked list of poses with calculated affinities (kcal/mol); the best-ranked pose was used for each compound, and top poses were inspected in PyMOL. Approximate dissociation constants were estimated from the docking scores using

$$K_D = \exp(\Delta G / RT),$$

with $RT \approx 0.593$ kcal/mol at 298 K. For example, for compound 2c, $\Delta G = -7.184$ kcal/mol gives $K_D = \exp(-12.11) = 5.5 \times 10^{-6}$ M ≈ 5.5 μM. Because docking scores are approximate, these K_D values are used only for relative comparison.

3. Results

3.1 Docking scores

All five active compounds docked with negative scores, while every pose of the prodrug 3a scored strongly positive (Table 3). Compound 2c had the best score (-7.184 kcal/mol; estimated $K_D \approx 5.5$ μM). The parent compound and 2a were effectively tied (-6.908 and -6.907 kcal/mol). The N⁶-methylated analogues scored weaker (2d, -6.089 ; 2e, -5.766 kcal/mol).

Table 3. SwissDock results compared with experimental activity.

Compound	Poses returned	Best ΔG (kcal/mol)	Est. K_D (μM)	IC_{50} (μM)	MERS-CoV EC_{50} (μM)	SI
1 (aristeromycin)	5	-6.908	8.73	1.32	2.0	> 25
2a (6'- β -F)	4	-6.907	8.74	0.37	0.20	3
2c (6',6'-diF)	3	-7.184	5.48	1.06	0.20	16
2d (N ⁶ -Me, 6'- β -F)	5	-6.089	34.72	4.39	> 50	n/a
2e (N ⁶ -Me, 6',6'-diF)	4	-5.766	59.86	0.76	12.5	> 4
3a (prodrug of 2c)	11	+22.043	not meaningful	> 100	9.3	> 5.3

$K_D = \exp(\Delta G/RT)$, $T = 298$ K, $RT = 0.593$ kcal/mol. 3a: all 11 poses scored between +22.0 and +24.7 kcal/mol, so no K_D is reported. IC_{50} is against SAH hydrolase; SI = selectivity index (CC_{50}/EC_{50}) as reported by Yoon et al. [1].

3.2 Binding poses

Visual inspection in PyMOL showed that the top-ranked poses of the active compounds occupied the substrate-binding pocket next to the NADH cofactor, in contact with pocket residues such as His55 and Asp131 (Figs. 7–8).

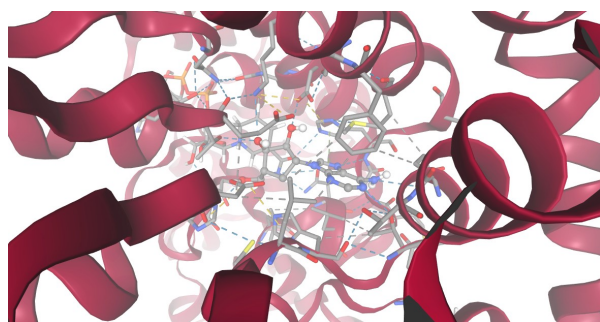
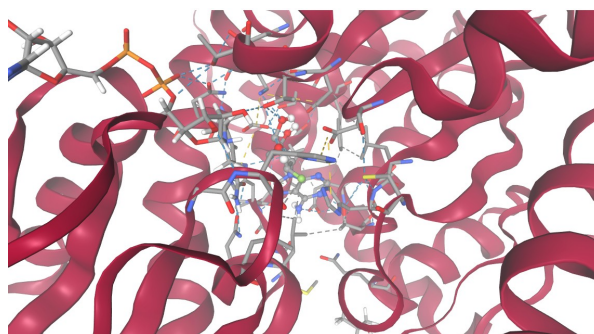
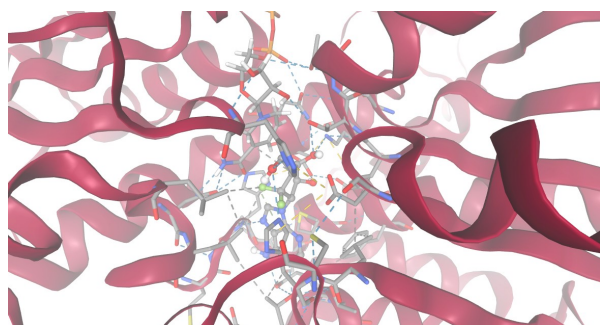
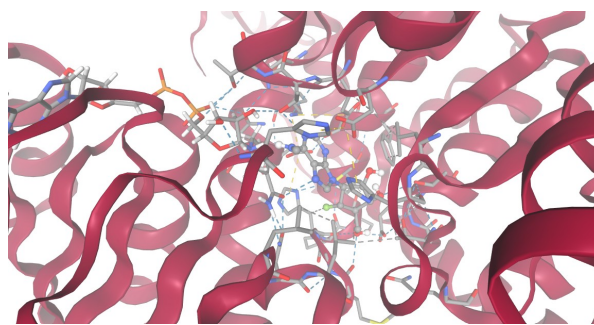
(a) **1**, aristeromycin (ΔG -6.908)(b) **2a** (ΔG -6.907)(c) **2c** (ΔG -7.184)(d) **2d** (ΔG -6.089)

Figure 7. Top-ranked docking poses of compounds **1**, **2a**, **2c** and **2d** in the AH CY active site (cartoon = protein; sticks = ligand and surrounding residues; dashed lines = predicted interactions).

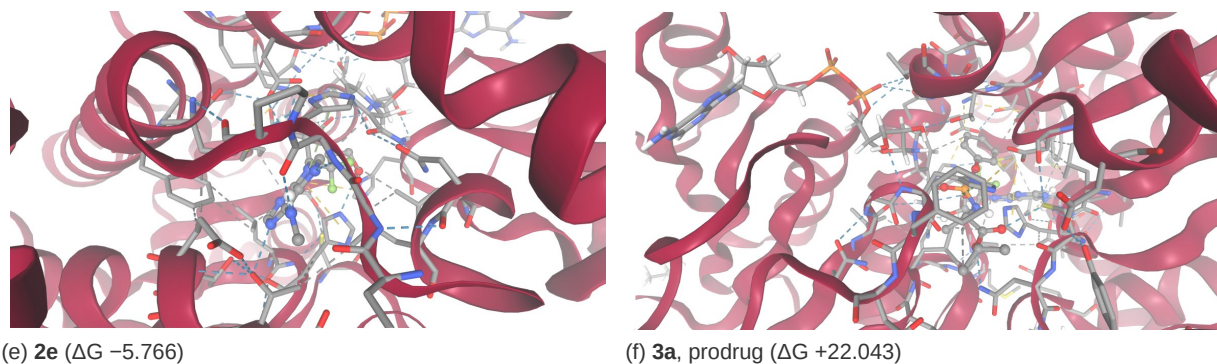


Figure 8. Top-ranked poses of 2e and of the phosphoramidate prodrug 3a. Panel (f) shows the least-unfavorable pose of 3a; every pose of this compound scored positive.

3.3 Docking score versus experimental potency

Plotting the best docking score against the experimental enzyme IC_{50} shows no clear rank relationship among the five active compounds (Spearman $\rho = -0.10$, $n = 5$; Fig. 9). The prodrug 3a is omitted from this plot because its IC_{50} is only known as a lower bound and its score reflects steric clash rather than binding.

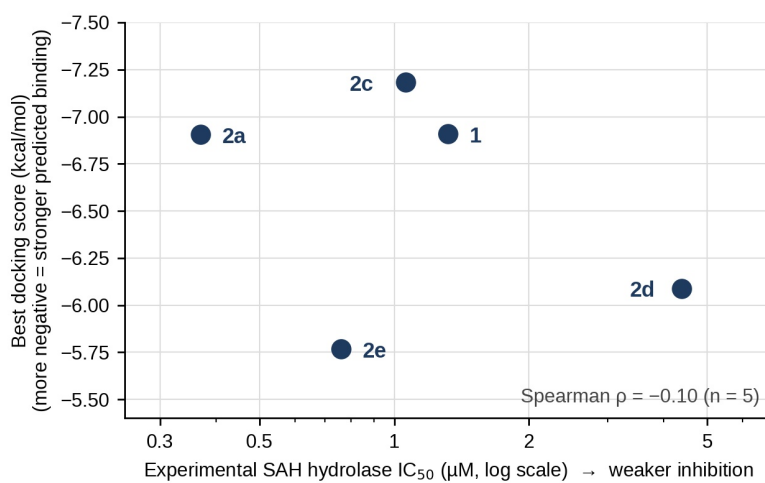


Figure 9. Best SwissDock score versus experimental SAH hydrolase IC_{50} for compounds 1, 2a, 2c, 2d and 2e.

4. Discussion

4.1 What the docking captured

The docking reproduced several qualitative features of the experimental data. All five active compounds were placed in the substrate pocket next to NADH with predicted affinities in a plausible range (-5.8 to -7.2 kcal/mol, $K_D \approx 5$ – $60 \mu M$). The top-scoring compound, 2c, is also the compound highlighted by Yoon et al. as the most promising broad-spectrum antiviral (MERS-CoV EC_{50} $0.20 \mu M$, SI 16). Compound 2d, which showed no antiviral activity, scored worse than the parent compound. And the prodrug 3a, the only compound with an experimental IC_{50} above $100 \mu M$, was the only one that could not be accommodated by the pocket.

4.2 Where docking and experiment diverge

The docking scores did not reproduce the potency ordering of the series. Compound 2a is the most potent enzyme inhibitor (IC_{50} 0.37 μ M) yet scored the same as the parent compound (−6.907 versus −6.908 kcal/mol), and 2e inhibits the enzyme more strongly than the parent (0.76 versus 1.32 μ M) yet received the weakest score of the five active compounds. Score differences of well under 1 kcal/mol, such as those among compounds 1, 2a and 2c, are smaller than the uncertainty typical of docking scoring functions, so the apparent superiority of 2c should be treated as a tentative observation rather than a resolved ranking. Several factors plausibly contribute:

- **Mechanism of inhibition.** Aristeromycin- and neplanocin-type nucleosides are generally understood to inhibit AHCY by mechanism-based inactivation: enzyme-bound NAD^+ oxidizes the 3'-hydroxyl, leaving NADH and a 3'-keto inhibitor, as captured in 1LI4 [2]. The receptor used here holds NADH, and ligands were docked in their native form, so the catalytic oxidation step and the tight-binding behavior reflected in IC_{50} values are not modeled.
- **Static receptor.** The receptor was rigid and docking does not capture protein dynamics or the ordered water network seen in high-resolution crystal structures.
- **Fluorine effects.** The influence of fluorine on the carbocyclic ring conformation, polarity and hydrogen bonding may not be well described by a general-purpose scoring function.
- **Different readouts.** IC_{50} and EC_{50} are functional measurements, not binding constants. EC_{50} also depends on cellular uptake and activation and, because these compounds were designed as dual-target agents, on inhibition of the viral RdRp, which docking to AHCY does not address [1].

4.3 The prodrug and the selectivity trade-off

Prodrug 3a scored between +22.0 and +24.7 kcal/mol across all 11 poses. Positive values of this size do not represent a physical binding energy; they indicate that the ligand cannot be placed in the rigid pocket without severe steric clashes. This is consistent with the bulky phosphoramidate group blocking access to the active site and with the experimental finding that 3a is inactive against the isolated enzyme ($IC_{50} > 100$ μ M) but antivirally active in cells (EC_{50} 9.3 μ M), as expected for a prodrug that must be processed intracellularly. The docking result agrees with this interpretation but does not by itself prove it.

The experimental selectivity indices also show that enzyme potency alone does not determine suitability: 2a is the strongest enzyme inhibitor but has a low SI (3), whereas 2c has a similar EC_{50} with a much better SI (16). Strong inhibition of host SAH hydrolase can disrupt cellular methylation and reduce cell viability, so the aim is a balanced inhibitor rather than the most potent one. Docking to AHCY alone cannot address cytotoxicity.

4.4 Choice of structural model

Using the experimental structure was appropriate for this project: it provides validated coordinates for the active site and the cofactor, whereas the ColabFold models had low confidence. Because AHCY is a well-conserved enzyme with many homologues, the low confidence most likely reflects the alignment depth and input used in this particular run rather than an intrinsically unpredictable fold. Repeating the prediction with the full-length human sequence (UniProt P23526) and comparing the result with 1LI4 would be a useful check.

5. Limitations and Future Work

- **Small data set and single docking run.** Five active compounds with scores that differ by small margins cannot support a statistical correlation. Repeated runs and a larger analogue set would help.
- **Protocol validation.** Re-docking the co-crystallized ligand (neplanocin) into the prepared receptor and reporting the RMSD would establish how well the protocol reproduces a known binding mode.
- **Receptor state.** Docking into an NAD⁺-bound structure, or docking 3'-keto forms of the ligands, would better represent the mechanism-based inhibition. Ensemble docking against other human AH CY structures (for example 3NJ4) would test the sensitivity to receptor conformation. Because AH CY functions as a tetramer, docking against a single chain may also omit contributions from neighboring subunits.
- **Dynamics and scoring.** Molecular dynamics refinement of top poses and rescoring with a more rigorous method (for example MM-GBSA) would provide a more reliable comparison than a single docking score.
- **Biological scope.** Host AH CY inhibition is only one proposed mode of action for these compounds; biochemical and cell-based assays remain essential.

6. Conclusion

Human SAH hydrolase is a rational target for indirect antiviral therapy against MERS-CoV, because inhibiting it is expected to disrupt the methylation reactions the virus needs for RNA capping. The experimental 1LI4 structure proved a more reliable docking receptor than the low-confidence AlphaFold model. Docking placed all active aristeromycin analogues in the AH CY substrate pocket, ranked 6',6'-difluoroaristeromycin (2c) highest, and identified the phosphoramidate prodrug 3a as unable to occupy the site, in line with published data [1]. However, the scores did not reproduce the potency ranking within the series, which highlights the limits of rigid-receptor docking for closely related and mechanism-based inhibitors. Docking is therefore best used here as a tool for pose analysis and coarse screening, alongside experimental assays. On the experimental evidence (EC₅₀ 0.20 μ M, SI 16), 2c remains a promising candidate for broad-spectrum antiviral development.

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